



Strategic growth option, uncertainty, and R&D investment



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ABSTRACT

This paper investigates the effect of uncertainty on R&D investment. We find that firms invest more in R&D when they face higher uncertainty, as measured by idiosyncratic return volatility. We further show that the effect is more pronounced for firms in more competitive industries as well as for firms whose products have less market power. Overall, our findings support the theory of strategic growth option in which firms under competition follow preemptive strategy when they face high uncertainty.

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1. Introduction

Does uncertainty impede or enhance R&D investment? Under the real option theory (Abel & Eberly, 1996; Abel, Eberly, & Pindyck, 1996; Dixit & Pindyck, 1994 and Pindyck, 1991), firms invest less when facing high uncertainty. This is because an increase in uncertainty leads to a higher value in the firms' option waiting rather than immediately undertaking irreversible and costly investments. In contrast, the theory of strategic growth option shows that under imperfect competition, uncertainty might encourage investment in a growth option. The reasoning behind this theory is that uncertainty can generate a growth option and delaying investments could leave the investment opportunity to other competitors while "immediate action may discourage entrants and enhance market share and profits" (Kulatilaka & Perotti, 1998), thus increasing competitive advantage in the future.

These theories rely on different assumptions. The real option theory assumes that a firm has a monopoly over an investment opportunity and that investment does not affect either product prices or market structure. On the other hand, the strategic growth option analyzes corporate investment under imperfect competition (Kulatilaka & Perotti, 1998). Specifically, when the product markets are not monopolistic, other potential competitors can seize the growth opportunities. In such markets, firms usually recognize that early investment, especially

R&D investment, is associated with greater opportunities to expand in the future¹.

Because the two theories predict opposite relations between uncertainty and investment in growth option, the empirical results are important. In this paper, we provide an answer by applying the strategic growth option theory to empirically examine the effect of uncertainty on R&D investment. We focus on R&D investment for the following reasons. First, R&D is the core business of firms (Porter and Millar, 1985). Second, in contrast to capital investment, R&D investment usually generates growth for firms in the future (Kumar & Li, 2016 and Lengnick-Hall, 1992) because of new technological progress and new physical capital that thus increase the productivity of physical investment (Lin, 2012). Third, while there is a great deal of literature on the relation between uncertainty and capital investment (e.g., Abel et al., 1996; Bulan, 2005; Dixit & Pindyck, 1994; Leahy & Whited, 1996 and Panousi & Papanikolaou, 2012), the effect of uncertainty on R&D investment is mixed (Czarnitzki & Toole, 2011; Minton & Schrand, 1999 and Stein & Stone, 2013) and even under-explored.

We use the idiosyncratic volatility in stock returns to represent uncertainty and find that firms invest more in R&D when they face higher

¹ For example, Nokia was the largest vendor of mobile phones in the 1990s and early 2000s. However, since its competitors took the advantage of and invested heavily in new phone technologies, Nokia lagged in the smartphone markets, which caused its share price to fall from \$40 in late 2007 to under \$2 in mid-2012. In 2013, its mobile business was then acquired by Microsoft.

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uncertainty. This finding supports the strategic growth option theory. This result holds under different regression specifications and after controlling for the firms' main characteristics such as Tobin's Q, cash holding ratio, return on equity, sales volume, dividend ratio, and age. This result also holds when we use other methods to estimate idiosyncratic volatility.

According to the strategic growth option theory, competition is the force behind the positive relation between uncertainty and investment. Therefore, we expect that the effect of uncertainty on R&D investment is more pronounced for firms in more competitive industries. Consistent with our prediction, we find that the positive relation between uncertainty and R&D investment is significantly stronger for the group of firms in more competitive industries.

In addition to market competition, the market power of a firm's product can also affect the relation between idiosyncratic return volatility and R&D investment. If the strategic growth option theory holds, we expect the preemptive strategy to be more important for the firms whose products have less market power because these firms face a greater pressure to survive. Consistent with this hypothesis, we find that the effect of idiosyncratic return volatility on R&D investment is stronger for firms whose products have less market power as measured by their sales volume.

To deal with the possible endogeneity between R&D investment and uncertainty, we control for the lag of the dependent variable - R&D investment. We further deal with this issue by using a 2SLS regression with instrumental variable. Following Panousi and Papanikolaou (2012), we use a firm's customer base concentration to instrument for its idiosyncratic return volatility. As discussed in Panousi and Papanikolaou (2012), customer base concentration should significantly correlate with idiosyncratic return volatility but is unlikely to directly affect R&D investment decisions. Our results strongly hold under this instrument regression.

Our paper is largely related to the literature on innovation and competitive advantage (e.g., Eberhart, Maxwell, & Siddique, 2004; Seru, 2014; Phillips & Zhdanov, 2013, and Vo, Le, & Ly, 2016), and the literature on the growth option (e.g., Chen, Zia, & Sun, 2016; Li, 2016 and Wagner, 2004). However, while these studies focus on the benefits of R&D investment or the relation between firms' characteristics and innovation or the value of the growth option, we investigate the effect of idiosyncratic return volatility on the decision to invest in R&D.

To the best of our knowledge, this is the first empirical paper that uses the theory of the strategic growth option to investigate the relation between uncertainty and R&D investment. Our paper also contributes to the literature by providing an explanation for the positive relation between uncertainty and R&D investment. Our paper further demonstrates the important role of a preemptive strategy under competition in corporate investment policies.

The rest of the paper is organized as follows. The next section reviews the related literature. Section 3 introduces the data collection, measurement, and statistics. We discuss the effect of uncertainty on R&D investment in Section 4. In Section 5 we examine the effect of competition on the relation between uncertainty and R&D investment. Section 6 presents the robustness tests of the results, and Section 7 concludes.

2. Literature review

In perfect capital markets, only the systematic component of risk is relevant for investment decisions since idiosyncratic risk can be diversifiable. However, a firm cannot fully diversify its operations in reality. As a result, both the theoretical models and empirical evidence show that uncertainty matters when making an investment decision. The literature on the real option theory argues that investments are at least irreversible and the adjustment costs are asymmetric and nonlinear with the costs of some input stocks. Under high adjustment costs, not investing allows firms to maintain their option to invest when

future business conditions become more attractive. When firms decide to make a capital investment decision, they exercise or "kill" their options to wait (McDonald & Stegel, 1986, and Pindyck, 1991). Moreover, Dixit and Pindyck (1994) show that the combination of uncertainty and irreversibility reduces the incentives for firms to immediately invest but increases the incentives to wait. Therefore, high uncertainty discourages investment while low uncertainty can increase the firms' motivation to invest. Thus, the relation between uncertainty and investment should be negative. Consistent with this theory, Leahy and Whited (1996) and Bulan (2005) use idiosyncratic return volatility to represent uncertainty and find that this volatility depresses capital investment.

Although the real option analysis has become a main stream in finance, it is often based on two specific assumptions (Kulatilaka & Perotti, 1998): (1) a firm has a monopoly over an investment opportunity, and (2) its action does not affect either the prices or the market structure. These assumptions can be reasonable when product markets are less competitive or monopolistic such as natural resource industries. However, they seem to be less realistic when the product markets become more competitive because other potential competitors can easily seize growth opportunities. In such markets, firms usually recognize that early investment, especially R&D investment, is associated with a greater ability to expand in the future.

Due to these limitations of the real option theory, the current literature has relaxed these assumptions to examine the effects of uncertainty on a firm's investment decision. Fudenberg and Tirole (1985) develop a model to show that firms can adopt a new technology sooner because of potential competitors. Grenadier (2002) shows that "exercise strategies cannot be determined separately but must be formed as part of a strategic equilibrium." Further, his model shows that the value of the option to wait is drastically eroded due to competition, and firms can invest at near the threshold of zero net present value. More importantly, Kulatilaka and Perotti (1998) develop a model for the strategic growth option to show that under imperfect competition, uncertainty can encourage investment in growth options. The underlying reason is that an initial investment is considered as the acquisition of growth opportunities relative to other competitors, which allows the firm to gain a competitive advantage. Therefore, when strategic investment has a significant preemptive effect, it results in higher market share, and thus greater ex post profits relative to the case of no investment in growth options. Thus, increased uncertainty can encourage investment in growth options when this strategic advantage is strong².

Focusing on R&D investment, Bloom (2007) develops a model to show that R&D investment is very persistent because the marginal effect of uncertainty on R&D is negative when firms increase their R&D investment but is positive when firms reduce their R&D investment. He calls for empirical research to examine this relation. Minton and Schrand (1999) calculate the cash flow volatility over a six-year period and examine its impact on the contemporaneous average corporate investments during the same period. They find that this cash flow volatility has a contemporaneous and negative relation to average capital investment, R&D, and advertising. Although the results are interesting, this method also shows some weaknesses. First, the firms' characteristics might be different before or after they make investment decisions, which could be problematic given that the average computation of these characteristics over a long period of time (six years) might not show their true relation. For example, firms with low idiosyncratic

² Although competitive advantage takes many forms, "a firm is said to have a competitive advantage when it is implementing a value creating strategy not simultaneously being implemented by any current or potential competitors" (Barney, 1991). Thus, if firms want to enhance their competitive advantage, they must have some strategies that are valuable and difficult to imitate (Barney, 1991 and Lengnick-Hall, 1992). As shown in the literature on strategic management, R&D investment does well to generate firms' competitive advantage, because beyond their benefits mentioned above, the outcomes of these investments are difficult to imitate (Lengnick-Hall, 1992). Importantly, this literature shows the firms' innovations as the cornerstones of their competitive advantage (Porter & Millar, 1985, Barney, 1991, and Lengnick-Hall, 1992).

risk invest more, but after they invest more, their idiosyncratic volatility increases (Duarte, Kogan, & Livdan, 2012). Second, the volatility might be endogenous with corporate investment policies.

Czarnitzki and Toole (2011 and 2013) examine a panel of German manufacturing firms and find that firms invest less in R&D when the absolute value of the sales of innovative products becomes more volatile. While the volatility in sales of innovative products can be a good proxy for uncertainty, it can be highly correlated and endogenous with R&D investment and might not reflect the uncertainty in total sales or a firm's idiosyncratic risk. Further, the authors measure R&D investments with the logarithm of absolute value of R&D, which might be dominated by firms with large R&D expenditures. Moreover, their estimated effects might be driven by omitted control variables, such as size, profitability, or cash flow.

In contrast, Jiang, Xu, and Yao (2009) report a positive correlation between idiosyncratic return volatility and R&D investment. However, they do not provide any suggestion about the causal effect between these two variables. Stein and Stone (2013) also find that uncertainty captured by implied volatility from equity options increases R&D investment. However, they assume that the reason is perhaps that the value of put options can offset the value of call options. If the value of put options can offset the value of call options, firms have more opportunities to sell capital assets at lower costs. Because the put option increases the incentive to invest more in capital (Abel et al., 1996), this reason may support the positive effect of uncertainty on capital investment, which seems to contradict the empirical evidence in the literature on the relation between uncertainty and investment (Abel et al., 1996; Berger, Ofek, & Swary, 1996 and Leahy & Whited, 1996).

Overall, the empirical papers on the relation between uncertainty and R&D investment are mainly based on the real option theory. This paper, on the other hand, applies the strategic growth option theory to analyze the effect of uncertainty on R&D investment.

3. Data, measures and descriptive statistics

3.1. Data and variables

We calculate our two main variables, uncertainty and R&D investment, using the Center for Research in Security Prices (CRSP) and Compustat data from 1985 to 2013. First, we define R&D investment as the ratio of R&D expenditure to total assets, both of which are collected from Compustat. If R&D expenditure is missing, we assume that it is zero for that year.

To measure uncertainty, we collect stock returns from the daily CRSP tapes for all ordinary common stocks (share code of 10 and 11). We use firms' daily stock returns to estimate annual idiosyncratic volatility that is the standard deviation of the residuals from the regression model of stock returns on market returns over a year. This annual idiosyncratic return volatility is our main proxy for uncertainty because it reflects the total uncertainty faced by a firm (Bulan, 2005). Idiosyncratic return volatility also reflects the volatility in the firm's profits and output price (Pindyck, 1991) as well as the volatility of both assets in place and growth opportunities (Berk, Richard, & Vasant, 1999 Carlson, Fisher, & Giammarino, 2004).

To robustly test our hypothesis, we compute the idiosyncratic return volatility as the standard deviation of the residuals from the Fama and French (1993) three-factor regression model. As mentioned in Ang, Hodrick, Xing, and Zhang (2006) and Jiang et al. (2009), this method is also broadly use in finance.

We calculate the firm's age from the CRSP data set where age is assigned the value of one in the year the firm is born and increases by one in each subsequent year.

For other control variables, we use Compustat to calculate the book value of equity, earnings on equity (ROE), cash holdings to total assets ratio, dividend to total assets ratio, and the logarithm of sale volume.

In addition, we use this data to compute the Herfindahl-Hirschman Index (HHI) based on sale volumes³.

We use two methods to calculate Tobin's Q. We first measure Tobin's Q as the ratio of the firm's total market value to its book value. This measure is mainly used to capture growth opportunities. To robustly test our hypothesis, we follow Salinger and Summers (1983) and define Tobin's Q alternatively as the ratio of a firm's market capitalization to its replacement cost of capital. We calculate market capitalization as the market value of equity plus the book value of liabilities minus deferred taxes. To calculate the firm's replacement cost (K), we initialize the first value of this variable (K_0) as gross PPE (PPEGT). We then construct the capital stock iteratively as $K_t = ((P_t/P_{t-1}) * K_{t-1} + CAPX_t)(1 - \delta_j)$, where P is the price deflator for fixed nonresidential investment from National Income and Product Accounts, CAPX is the capital expenditure, and δ_j is the book depreciation rate at the three-digit SIC level. We calculate $\delta_j = 2/L_j$ where L_j is the useful life of a capital good that is computed as $L_j = \frac{1}{N_j} \sum_{i \in j} \frac{PPE_{t-1} + I_t}{D_t}$ in which D is depreciation (DP).

We use the Compustat segment files to compute customer base concentration, which is the instrumental variable for uncertainty. Following Panousi and Papanikolaou (2012), we define this measure as the Herfindahl-Hirschman concentration index of a firm's sales across its customers.

We eliminate any observations with total assets, market capitalization, and book value of less than \$1 million, or with missing total sale volumes. Financial, utility, and other regulated companies are also excluded from our sample⁴. Further, any observation without data on idiosyncratic return volatility is eliminated from the sample. Finally, to reduce the effects of outliers, we winsorize all variables at the 1st and 99th percentile levels. Our final sample consists of 90,650 firm-year observations.

3.2. Descriptive statistics

Table 1 shows the descriptive statistics of the sample firms. The average of a firm's R&D investments to total assets ratio is 5.30% while this value at the 25th and 75th percentiles is zero and 6.50%, respectively. This evidence shows that R&D investments are clustered in some groups of firms, which is consistent with the current literature (e.g., Chen et al., 2001, and Eberhart et al., 2004) that R&D is more concentrated among firms in high-tech industries.

Panel A of Table 1 also shows that the idiosyncratic return volatility ranges from 0.009 to 0.119. The average is 0.036 while the value at the 25th and 75th percentiles is 0.021 and 0.046 respectively. Tobin's Q, which is the ratio of a firm's market value to its book value, is from 0.563 to 9.669, and averages 1.929. This panel also reports the statistics for the other main variables in our paper. For example, the average cash holding ratio for firms is 0.181, while the average return on equity is 0.018. Similarly, the average logarithm of total assets is 5.126 while the median is 4.988. The average logarithm of sales volume is 5.029 while the median is 5.006.

Next, we sort all firms in our sample into four groups for every year based on their industry and idiosyncratic return volatility. The results in panel B report the statistics for firms in each group.

Panel B shows that idiosyncratic return volatility monotonically increases with R&D investment. The average idiosyncratic return volatility for firms in the first group is 0.020, while this figure for the last group is 0.063. At the same time, the average R&D investment ratio for the firms in group 1 is 0.034, and this figure for firms in group 4 is 0.067. This result supports our hypothesis that firms with high idiosyncratic return volatility invest more in R&D. Our results in this panel are consistent with those reported in the literature (Jiang et al., 2009).

³ More detailed calculations of all variables can be found in Appendix A.

⁴ The industries are taken from Barclay and Clifford (1995).

Table 1
Descriptive statistics.

Panel A: descriptive statistics							
Variable	Maximum	Minimum	Mean	Median	25th	75th	N
VOL	0.119	0.009	0.036	0.031	0.021	0.046	90,650
RDAT	0.564	0.000	0.053	0.001	0.000	0.065	90,650
Q	9.669	0.563	1.929	1.418	1.057	2.162	90,650
CASH	0.870	0.000	0.181	0.095	0.026	0.265	90,650
ROE	1.428	−2.261	0.018	0.083	−0.058	0.178	90,650
DIV	12.612	−0.374	0.010	0.000	0.000	0.010	90,650
LAGE	4.489	0.000	2.296	2.398	1.609	3.045	90,650
LAT	10.402	0.909	5.126	4.988	3.583	6.556	90,650
LSALE	10.334	−0.865	5.029	5.006	3.503	6.567	90,650
HHI	1.000	0.029	0.167	0.123	0.075	0.205	90,650
Panel B: descriptive statistics by idiosyncratic volatility group.							
VOL group	Low						High
	1	2	3	4			
VOL	0.020	0.029	0.040	0.063			
RDAT	0.034	0.054	0.059	0.067			
Q	1.941	1.983	1.923	1.880			
CASH	0.142	0.186	0.194	0.199			
ROE	0.133	0.078	−0.001	−0.174			
DIV	0.016	0.009	0.007	0.006			
LAGE	2.702	2.248	2.053	1.951			
LAT	6.390	5.352	4.545	3.526			
LSALE	6.460	5.356	4.538	3.466			
HHI	0.182	0.157	0.170	0.148			
N	23,332	20,142	21,692	18,896			

This table reports the descriptive statistics of all firms in our sample during the period from 1985 to 2013. The VOL is the idiosyncratic return volatility. RDAT is the ratio of R&D expenditures to total assets. The Q is the Tobin's Q, a firm's total market value to its book value. The CASH is the fraction of cash holding to total assets, ROE is the return on equity, DIV is the dividend payout to total assets, LAGE is the logarithm of the firm's age, LAT is the logarithm of total assets, LSALE is the logarithm of the firm's sales volume, and HHI is the Herfindahl-Hirschman Index (HHI) calculated from the sales volume. All variables are defined in Appendix A. Panel A reports the descriptive statistics of the entire sample, and Panel B shows the average statistics of firms in each group based on idiosyncratic return volatility.

The results in panel B also demonstrate that the idiosyncratic return volatility is negatively correlated with the firm's age, total assets, and sales volume. The average age in the lowest idiosyncratic return volatility group (group 1) is 15 years while this figure in the highest idiosyncratic return volatility group (group 4) is 7 years. Similarly, the firms in group 1 on average have total assets of 6.390 (in logarithm), and the firms in group 4 have total assets of 3.526 (in logarithm). The average logarithm of sales volumes also decreases from 6.460 for group 1 to 3.446 for group 4. These results show that large or old firms have less idiosyncratic return volatility. These firms also invest less in R&D. These findings are consistent with the findings in the current literature (e.g., Phillips & Zhdanov, 2013 and Vo et al., 2016).

4. The main results

4.1. Base-line regression specification

The goal of this paper is to examine the effect of uncertainty on R&D investment. To explore this effect, following the Tobin's Q theory, we start with the following regression model:

$$RDAT_{i,t+1} = \beta_0 + \beta_1 Q_{i,t} + \beta_2 VOL_{i,t} + f_i + y_t + \varepsilon_{i,t} \quad (1)$$

where t denotes time; RDAT is the ratio of R&D expenditures to total assets; Q is Tobin's Q, which is the ratio of a firm's market value to its book value; VOL is the idiosyncratic return volatility, f_i is a firm dummy variable, and y_t is a year dummy.

The orthodox theory of investment developed by Tobin (1969) shows that firms with high growth opportunities tend to invest more. Therefore, these growth opportunities should be controlled when examining the effect of idiosyncratic return volatility on R&D investment. In addition, since the propensity to invest might be different for

different firms and/or for firms in different industries, we control for firm or industry fixed effects. We further control for time effects because both volatility and a firm's R&D investments are related to business cycles and macroeconomic variables.

We expand the model by adding other control variables that potentially affect a firm's R&D investment. We include the cash holding ratio, sales volume, and profitability because these variables are significantly correlated with a firm's innovation (Brown, Fazzari, & Petersen, 2009; Hall, Jaffe, & Trajtenberg, 2005). Further, we include the dividend ratio because this factor can affect a firm's investment policy. We use age to control for the effect a firm's life cycle and profitability uncertainty on the relation between idiosyncratic return volatility and R&D investment (Pastor & Veronesi, 2003).

Because endogeneity might exist in the relation between uncertainty and R&D investment, we follow the current literature (e.g., Bloom, Bond, & Reenen, 2007; Bulan, 2005) to control for the lag in R&D investment. This lag can be used as an internal instrument. To be specific, our regression model is as follows:

$$RDAT_{i,t+1} = \beta_0 + \beta_1 Q_{i,t} + \beta_2 VOL_{i,t} + \beta_3 RDAT_{i,t} + \beta_4 FIRM_CHAR_{i,t} + f_i + y_t + \varepsilon_{i,t} \quad (2)$$

where FIRM_CHAR is a set of the firm's characteristics as described.

4.2. The results

Table 2 shows the results from regression (2). The results from this table show that, in general, the idiosyncratic return volatility has a significantly positive correlation with R&D investment. Specifically, the coefficient for VOL in the regression on R&D investment (model 1) is 0.162, which is statistically significant at the 1% level. This coefficient means that when idiosyncratic volatility increases by 1%, firms increase R&D investment by 16.2%, economically significant. This result holds when we control for other characteristics (columns 4 and 5).

Table 2
Idiosyncratic volatility and R&D investment.

	(1)	(2)	(3)	(4)	(5)
	RDAT _{i,t+1}	RDAT _{i,t+1}	RDAT _{i,t+1}	RDAT _{i,t+1}	RDAT _{i,t+1}
VOL _{i,t}	0.162*** (0.000)	0.114*** (0.000)	0.135*** (0.000)	0.039*** (0.005)	0.069*** (0.000)
Q _{i,t}	0.009*** (0.000)	0.005*** (0.000)	0.003*** (0.000)	0.005*** (0.000)	0.003*** (0.000)
RDAT _{i,t}		0.348*** (0.000)	0.730*** (0.000)	0.343*** (0.000)	0.715*** (0.000)
CASH _{i,t}				−0.013*** (0.000)	0.021*** (0.000)
ROE _{i,t}				0.000 (0.596)	0.000 (0.581)
DIV _{i,t}				−0.013*** (0.000)	−0.012*** (0.000)
LAGE _{i,t}				0.005*** (0.000)	0.002*** (0.000)
LSALE _{i,t}				−0.008*** (0.000)	−0.002*** (0.000)
HHI _{j,t}				0.000 (0.928)	−0.002 (0.171)
Intercept	0.030*** (0.000)	0.018*** (0.000)	−0.000 (0.532)	0.048*** (0.000)	0.001 (0.387)
Fixed effects	F, T	F, T	Ind, T	F, T	Ind, T
N	90,650	90,650	90,650	90,650	90,650
Adj-R ²	0.1777	0.7572	0.7823	0.6856	0.7849

This table reports the results from the regression model of R&D investment on idiosyncratic volatility and other control variables: $RDAT_{i,t+1} = \beta_0 + \beta_1 Q_{i,t} + \beta_2 VOL_{i,t} + \beta_3 RDAT_{i,t} + \beta_4 FIRM_CHAR_{i,t} + f_i + y_t + \varepsilon_{i,t}$ (2). The RDAT is the ratio of R&D investment to total assets, Q is the Tobin's Q that is the ratio of a firm's market value to its book value, VOL is the firm's idiosyncratic volatility, and FIRM_CHAR is the firm's characteristics that potentially affect the firm's capital investment policies, f_i is a firm/industry dummy, and y_t is year dummy. All variables are defined in Appendix A. The sample is the period from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1% respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

Consistent with Tobin's Q, firms with high growth opportunities invest more in R&D. The p-values of the coefficients for Q are smaller than 1% that indicates a strong relation between Tobin's Q and R&D investment. In contrast, the effect of the cash holding ratio on R&D investment depends on the regression specification. The cash holding ratio is positively correlated with R&D investment under the regression model with industry fixed effects but negatively related to R&D investment when the firm fixed effect is used.

Different from the effect of cash holding on R&D investment, the effects of other firm characteristics on R&D investments are consistent among regression models. For example, firms with large sale volumes and firms that pay dividends invest less in R&D, consistent with the findings in the current literature (Phillips & Zhdanov, 2013). In addition, the results from Table 2 show that the firm's age is positively correlated with R&D investment.

Overall, the results from Table 2 support the prediction of the strategic growth option theory that firms invest more in R&D when they face high uncertainty.

5. Competition, uncertainty, and R&D investment

We have shown that our test results support the strategic growth option theory's prediction that firms invest more in R&D when they face high uncertainty. The driving force behind the positive effect of uncertainty on investment, according to the strategic growth option theory, is competition. Therefore, in this section, we examine the relation between uncertainty and R&D investment under different competition environments and also for firms with different competitive positions on the product markets.

5.1. Product market competition, uncertainty, and R&D investment

This subsection examines the effects of product market competition on the relation between uncertainty and R&D investment. Under high competition, firms are under greater pressure to survive and have more incentives to increase their competitive advantage. Further, the value of the option to wait can be easily eroded. Thus, if the strategic growth option theory holds, we expect that the effect of uncertainty on R&D investment is greater for firms in more competitive industries.

We measure product market competition for each industry using the Herfindahl-Hirschman index (HHI), defined as:

$$HHI(\text{Sale})_j = \sum_{i=1}^J s_{ij}^2$$

where s_{ij} is the proportion of sales of firm i in industry j , and J is the number of firms in industry j . We calculate HHI for each year and for each industry by using the three-digit SIC code to define the industry, and then average these values over the past three years. A large value for this index indicates that the market is concentrated in a group of firms, while a small value indicates that the market is shared by many competing firms⁵.

To examine the effect of competition on the relation between idiosyncratic volatility and R&D investments, every year we sort all firms into two equal groups based on their industry concentration index. We then use regression (2) to regress R&D investment on idiosyncratic volatility and other firm characteristics for each group. The results are reported in Table 3.

Table 3 shows that idiosyncratic return volatility is positively correlated with future R&D investment. However, the estimated coefficients for idiosyncratic return volatility are statistically insignificant for firms

Table 3
Industry competition, idiosyncratic volatility, and R&D investment.

	Concentrated industries		Competitive industries		Whole sample
	(1) RDAT _{i,t+1}	(2) RDAT _{i,t+1}	(3) RDAT _{i,t+1}	(4) RDAT _{i,t+1}	(5) RDAT _{i,t+1}
VOL _{i,t}	0.014 (0.400)	0.014 (0.400)	0.069*** (0.009)	0.069*** (0.009)	0.067*** (0.000)
VOLxHHIG _{i,t}					−0.052*** (0.000)
Q _{i,t}	0.003*** (0.000)	0.003*** (0.000)	0.007*** (0.000)	0.007*** (0.000)	0.005*** (0.000)
RDAT _{i,t}	0.386*** (0.000)	0.386*** (0.000)	0.309*** (0.000)	0.309*** (0.000)	0.342*** (0.000)
CASH _{i,t}	−0.007** (0.015)	−0.007** (0.015)	−0.017*** (0.000)	−0.017*** (0.000)	−0.013*** (0.000)
ROE _{i,t}	−0.000 (0.704)	−0.000 (0.704)	0.001 (0.303)	0.001 (0.303)	0.000 (0.581)
DIV _{i,t}	−0.003 (0.417)	−0.003 (0.417)	−0.016*** (0.000)	−0.016*** (0.000)	−0.014*** (0.000)
LAGE _{i,t}	0.002*** (0.000)	0.002*** (0.000)	0.006*** (0.000)	0.006*** (0.000)	0.005*** (0.000)
LSALE _{i,t}	−0.004*** (0.000)	−0.004*** (0.000)	−0.012*** (0.000)	−0.012*** (0.000)	−0.008*** (0.000)
HHI _{j,t}	0.002 (0.112)	0.002 (0.112)	−0.031 (0.103)	−0.031 (0.103)	0.002 (0.189)
Intercept	0.024*** (0.000)	0.024*** (0.000)	0.075*** (0.000)	0.075*** (0.000)	0.048*** (0.000)
Fixed effects	F,T	F,T	F,T	F,T	F,T
N	44,097	44,097	46,553	46,553	90,650
Adj-R ²	0.7202	0.7202	0.6284	0.6284	0.6869

This table reports the results from the regression of R&D investment (RDAT) on idiosyncratic return volatility (VOL) and other control variables. The HHIG is a dummy variable that is equal to zero if firms are classified as in competitive industries and one otherwise. Models (1) and (2) are for the sample of firms in concentrated industries; models (3) and (4) are for the sample of firms in competitive industries. All other variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1% respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

in the concentrated industries (columns 1 and 2). In contrast, for firms in more competitive industries, the coefficients for idiosyncratic return volatility are highly significant (columns 3 and 4).

In the last column of Table 3 we add the interaction term for the idiosyncratic return volatility and the competition group dummy (HHIG) to the regression. In this regression, the HHIG is assigned a value of one for firms in the concentrated industries, and zero otherwise. If the effect of uncertainty on R&D investment is stronger for firms in competitive industries we expect the interaction coefficient to be negative. Consistent with our prediction, the interaction term is negative and statistically significant at the 1% level.

5.2. Product market power, uncertainty, and R&D investment

If the incentive to take the preemptive strategy under a competitive environment is the main driver of the positive effect of uncertainty on R&D investment, we should expect that this effect is more pronounced for firms with smaller product market power. This is because firms with less market power have greater pressure to survive, so they have higher incentives to take advantage of future growth opportunities. Thus, we expect that the effect of uncertainty on R&D investment will be more pronounced for these firms.

To test this hypothesis, we use the logarithm of sales volume as a proxy for a firm's competition power and add the interaction term of this variable and idiosyncratic return volatility into the regression model (2)⁶. Since firms with lower market power tend to have a higher

⁵ Some other common methods to measure the HHI are to use market capitalization or total assets to calculate market share. Because these measures are highly correlated with each other, we focus on the HHI index calculated by sales. The results are qualitatively the same when we use the HHI measured by total assets or market capitalization.

⁶ Our results are qualitatively the same if we use a firm's market share within an industry to proxy for its product market power.

incentive to invest more in R&D, we expect this interaction term to be negative. We present the regression results in Table 4.

Table 4 shows that the coefficient for the interaction terms for volatility and the logarithm of sales volume is negative, as expected, and statistically significant at the 1% level. The results confirm that firms with less market power tend to invest more in R&D when they face higher uncertainty.

6. Robustness checks

6.1. Addressing endogeneity

An important issue that arises when examining the effect of uncertainty on R&D investment is endogeneity. On the one hand, uncertainty can affect R&D investment. On the other hand, firms with a greater proportion of R&D investment can have higher uncertainty since R&D investments are usually associated with new and untested technologies (Chan et al., 2001 and Lin, 2012). We have partially addressed the issue by controlling for the lag in R&D investment. However, the endogeneity problem may still exist. In this section, we deal with this issue by using 2SLS regression method with external instrumental variable.

We follow Panousi and Papanikolaou (2012) to use a firm's customer base concentration as an external instrument variable for idiosyncratic return volatility⁷. The concentration in a company's customer base can be an instrument because it significantly affects idiosyncratic return volatility. The reason is that a firm having a low customer concentration should have a high ability to diversify its customers' idiosyncratic demand shocks, leading to a low idiosyncratic return volatility. Further, customer base concentration is unlikely to directly affect R&D investment.

We define customer base concentration (HCON) as the Herfindahl-Hirschman concentration index of a firm's sales across customers. To be specific, this variable is calculated as:

$$HCON_{i,t} = \sum_p \left(\frac{Sales_{i,p,t}}{Sales_{i,t}} \right)^2$$

where $Sales_{i,p,t}$ is the reported sales of firm i to customer p in year t , and $Sales_{i,t}$ is the total sales volume of firm i in year t . A higher HCON value implies that the firm's customer base is more concentrated.

This variable has two properties: (1) it is not much varied over time, and (2) its identification mainly shows the cross-sectional dimension of the panel. Thus, we follow Panousi and Papanikolaou (2012) to drop the firm fixed effects in the tests but remain industry fixed effects in this subsection. We report the 2SLS regression results in Table 5.

The first column of Table 5 shows the results in the first-stage regression of VOL on HCON and other control variables. As expected, the coefficient of HCON is positive (0.001) and statistically highly significant ($p < 1\%$). This result means that a firm's idiosyncratic return volatility increases when a firm has a higher concentration in its customer base. We then use Cragg-Donald Wald F-statistic and Stock and Yogo test to test for weak instruments. The results (not reported) show that the null hypothesis of weak instruments is rejected, which means that customer base concentration is a valid instrument for idiosyncratic volatility.

The last column of Table 5 report the results from the second-stage regression. This column shows that the effect of idiosyncratic return volatility on R&D remains significant (p -value $< 1\%$) when instrumental

Table 4

Product market power, idiosyncratic volatility, and R&D investment.

	(1) RDAT _{i,t+1}	(2) RDAT _{i,t+1}	(3) RDAT _{i,t+1}
VOL _{i,t}	0.125*** (0.000)	0.108*** (0.002)	0.153*** (0.000)
VOLxLSALE _{i,t}	−0.022*** (0.002)	−0.020*** (0.005)	−0.025*** (0.000)
Q _{i,t}	0.005*** (0.000)	0.005*** (0.000)	0.003*** (0.000)
RDAT _{i,t}	0.340*** (0.000)	0.342*** (0.000)	0.715*** (0.000)
CASH _{i,t}		−0.013*** (0.000)	0.021*** (0.000)
ROE _{i,t}		0.000 (0.576)	0.001 (0.416)
DIV _{i,t}		−0.014*** (0.000)	−0.013*** (0.000)
LAGE _{i,t}		0.005*** (0.000)	0.002*** (0.000)
LSALE _{i,t}	−0.006*** (0.000)	−0.008*** (0.000)	−0.001*** (0.000)
HHI _{j,t}		0.000 (0.957)	−0.002 (0.128)
Intercept	0.047*** (0.000)	0.048*** (0.000)	−0.001 (0.266)
Fixed effects	F, T	F, T	Ind, T
N	90,650	90,650	90,650
Adj-R ²	0.7016	0.6828	0.7857

This table reports the results from the regression of R&D investment (RDAT) on idiosyncratic return volatility (VOL), the interaction of VOL and sales (LSALE), and other control variables. All variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1% respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

variable is used. This result confirms our finding that firms will invest more in R&D when they face high idiosyncratic volatility.

6.2. Alternative measures of Tobin's Q

The orthodox theory of investment developed by Tobin (1969) compares the capitalized marginal investment to its purchase cost. In reality, the ownership of investment is not traded in the market, thus the replacement cost is not directly observed. If Tobin's Q is an imperfect measure of investment opportunities, an omitted variable bias in the empirical estimate could exist, and some latent variables affecting both uncertainty and investments might also exist. Therefore, in this section we follow Salinger and Summers (1983) and estimate Tobin's Q as the ratio of a firm's total market value to its replacement cost of capital.

We use regression (2) with the alternative Tobin's Q in the logarithm to investigate the effect of idiosyncratic return volatility on R&D investment. The results are reported in Table 6.

Table 6 shows that in two of the three regressions, the idiosyncratic return volatility is positively correlated with R&D investment, confirming our previous test results. However, the effect of the alternative Tobin's Q on R&D investment varies with regression specifications. After controlling for some main firm's characteristics, this Tobin's Q is negatively related to R&D investment.

6.3. Non-high tech industries

To reduce the endogeneity between uncertainty and R&D investment, we narrow our sample to the firms in non-high-tech industries. This is to avoid the strong idiosyncratic volatility of firms in the high-tech industries. We follow Fama and French (1997) and define high-tech industries as industries with the following four-digit SIC codes:

⁷ In the earlier version of the paper we use two other external instruments for idiosyncratic return volatility (the exposures to the uncertainty in the US regulations and to the uncertainty in the US foreign exchange rates) and obtain qualitatively the same results.

Table 5
Addressing endogeneity issue.

	First stage VOL _{i,t}	Second stage RDAT _{i,t+1}
HCON _{i,t}	0.001*** (0.000)	
$\widehat{VOL}_{i,t}$		1.772*** (0.009)
Q _{i,t}	−0.001*** (0.000)	0.005*** (0.000)
RDAT _{i,t}	0.004*** (0.000)	0.708*** (0.000)
CASH _{i,t}	−0.009*** (0.000)	0.037*** (0.000)
ROE _{i,t}	−0.007*** (0.000)	0.012** (0.017)
DIV _{i,t}	−0.005*** (0.000)	−0.003 (0.478)
LAGE _{i,t}	−0.002*** (0.000)	0.006*** (0.000)
LSALE _{i,t}	−0.005*** (0.000)	0.007* (0.055)
HHI _{j,t}	0.003*** (0.000)	−0.006** (0.033)
Intercept	0.073*** (0.000)	−0.138** (0.016)
Fixed effects	Ind, T	Ind, T
Instrument		HCON
N	90,650	90,650
Adj-R ²	0.4826	0.7079

This table reports the results from the first and second stages of the 2SLS regression of R&D investment (RDAT) on idiosyncratic volatility (VOL) and other control variables. The external instrumental variable is the firm's customer base concentration (HCON). All variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1% respectively. T denotes year fixed effect, and Ind represents industry fixed effect.

3570 to 3579, 3600 to 3629, 3640 to 3646, 3648 to 3649, 3660 to 3692, 3695 to 3699, 4800 to 4899, 7370 to 7373, and 7375. After excluding the firms in these industries, we use regression model 2 and obtain

Table 6
Alternative Tobin's Q and R&D investment measures.

	(1) RDAT _{i,t+1}	(2) RDAT _{i,t+1}	(3) RDAT _{i,t+1}
VOL _{i,t}	0.123*** (0.000)	0.000 (0.990)	0.057*** (0.000)
LOGQ _{i,t}	0.000 (0.992)	−0.004*** (0.000)	−0.002*** (0.000)
RDAT _{i,t}		0.422*** (0.000)	0.792*** (0.000)
CASH _{i,t}		0.006** (0.033)	0.030*** (0.000)
ROE _{i,t}		0.000 (0.981)	−0.001 (0.104)
DIV _{i,t}		0.004 (0.192)	−0.006** (0.030)
LAGE _{i,t}		−0.000 (0.858)	0.000 (0.999)
LOGK _{i,t}		−0.006*** (0.000)	−0.000* (0.066)
HHI _{j,t}		0.002 (0.150)	0.001 (0.264)
Intercept	0.047*** (0.000)	0.053*** (0.000)	0.003*** (0.004)
Fixed effects	F, T	F, T	Ind, T
N	73,266	73,266	73,266
Adj-R ²	0.0109	0.2083	0.8104

This table reports the results from the regression of R&D investment (RDAT) on idiosyncratic volatility (VOL), an alternative Tobin's Q (LOGQ), and other control variables. The LOGQ is the logarithm of the firm's market value to its replacement costs. All other variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

qualitatively similar results for the positive effect of uncertainty on R&D investment (Table 7).

6.4. Changes in volatility and R&D investment

In the previous sections, we focus on the effect of idiosyncratic volatility on R&D investment. Since R&D investment and idiosyncratic return volatility might be persistent and endogenous, we test the robustness of our hypothesis that firms will invest more in R&D when they face higher uncertainty by examining the effect of the change in idiosyncratic return volatility on the change in R&D. We modify regression 2 to reflect this change. Our model is now as follows:

$$\Delta RDAT_{i,t+1} = \beta_0 + \beta_1 Q_{i,t} + \beta_2 \Delta VOL_{i,t} (VOL_{i,t}) + \beta_3 RDAT_{i,t} + \beta_4 FIRM_CHAR_{i,t} + f_i + y_t + \varepsilon_{i,t} \quad (3)$$

where $\Delta RDAT$ is the change in R&D investment, ΔVOL is the change in idiosyncratic volatility, and the other variables are as defined in regression 2.

If our hypothesis holds, we expect to see the positive effect of the change in idiosyncratic volatility on the change in R&D investment. This effect means that the coefficients for the change in volatility (β_2) are significantly positive. The results of this regression are reported in Table 8.

As we expect, the change in idiosyncratic volatility positively related to the change in R&D investment. Consistent with the results from the previous sections, this result shows that when idiosyncratic volatility increases, firms invest more in R&D.

6.5. Alternative measure of idiosyncratic return volatility

We primarily use the idiosyncratic volatility as the regression residuals of stock returns by using the CAPM regression model and investigate its impact on R&D investment. However, idiosyncratic volatility

Table 7
Non-high tech firms.

	(1) RDAT _{i,t+1}	(2) RDAT _{i,t+1}	(3) RDAT _{i,t+1}	(4) RDAT _{i,t+1}
VOL _{i,t}	0.108*** (0.000)	0.069*** (0.000)	0.014 (0.354)	0.024** (0.041)
Q _{i,t}	0.008*** (0.000)	0.005*** (0.000)	0.005*** (0.000)	0.003*** (0.000)
RDAT _{i,t}		0.385*** (0.000)	0.380*** (0.000)	0.738*** (0.000)
CASH _{i,t}			−0.010*** (0.001)	0.019*** (0.000)
ROE _{i,t}			0.001 (0.133)	0.001 (0.110)
DIV _{i,t}			−0.012*** (0.000)	−0.011*** (0.000)
LAGE _{i,t}			0.003*** (0.000)	0.002*** (0.000)
LSALE _{i,t}			−0.006*** (0.000)	−0.001*** (0.000)
HHI _{j,t}			0.001 (0.420)	0.000 (0.775)
Intercept	0.017*** (0.000)	0.010*** (0.000)	0.034*** (0.000)	0.000 (0.922)
Fixed effects	F, T	F, T	F, T	Ind, T
N	66,192	66,192	66,192	66,192
Adj-R ²	0.2037	0.7986	0.7494	0.819

This table reports the results from the regression model of R&D investment (RDAT) on idiosyncratic volatility (VOL) and other control variables for firms in non-high tech industries. All variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

Table 8
Change in idiosyncratic volatility and change in R&D investment.

	(1) $\Delta \text{RDAT}_{i,t+1}$	(2) $\Delta \text{RDAT}_{i,t+1}$	(3) $\Delta \text{RDAT}_{i,t+1}$
$\text{VOL}_{i,t}$	0.034** (0.027)		
$\Delta \text{VOL}_{i,t}$		0.013 (0.236)	0.029** (0.019)
$Q_{i,t}$	0.005*** (0.000)	0.005*** (0.000)	0.003*** (0.000)
$\text{RDAT}_{i,t}$	−0.619*** (0.000)	−0.619*** (0.000)	−0.246*** (0.000)
$\text{CASH}_{i,t}$	−0.009*** (0.000)	−0.009*** (0.000)	0.023*** (0.000)
$\text{ROE}_{i,t}$	0.001 (0.211)	0.001 (0.267)	0.001* (0.070)
$\text{DIV}_{i,t}$	−0.011*** (0.000)	−0.011*** (0.000)	−0.011*** (0.000)
$\text{LAGE}_{i,t}$	0.002*** (0.002)	0.002*** (0.002)	0.000 (0.422)
$\text{LSALE}_{i,t}$	−0.008*** (0.000)	−0.008*** (0.000)	−0.002*** (0.000)
$\text{HHI}_{j,t}$	0.000 (0.789)	0.001 (0.740)	0.000 (0.762)
Intercept	0.052*** (0.000)	0.054*** (0.000)	0.007*** (0.000)
Fixed effects	F, T	F, T	Ind, T
N	86,545	86,532	86,532
Adj-R ²	0.1024	0.1023	0.1473

This table reports the results from the regression model of change in R&D investment on the change in idiosyncratic volatility and other control variables: $\Delta \text{RDAT}_{i,t+1} = \beta_0 + \beta_1 Q_{i,t} + \beta_2 \Delta \text{VOL}_{i,t} + \beta_3 \text{RDAT}_{i,t} + \beta_4 \text{FIRM_CHAR}_{i,t} + f_i + y_t + \varepsilon_{i,t}$ (3) where ΔRDAT is the change in R&D investment, and ΔVOL is the change in the firm's idiosyncratic volatility. f_i is a firm/industry dummy and y_t is year dummy. All variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

can be measured in different ways. In this section, we estimate idiosyncratic return volatility by using the Fama and French (1993) three-factor regression model (VOLFF3). This method is also widely used in the finance literature (e.g., Ang et al., 2006 and Jiang et al., 2009.) To be specific, VOLFF3 is the standard deviation of the residuals ($\varepsilon_{i,t}$) from the following regression:

$$r_{i,t} = \alpha + \beta_{m,t}(r_{m,t}) + \beta_{\text{SMB},t}(\text{SMB}_t) + \beta_{\text{HML},t}(\text{HML}_t) + \varepsilon_{i,t} \quad (4)$$

where $r_{i,t}$ is the daily excess return on stock i , $r_{m,t}$ is the daily excess market return, SMB_t and HML_t are the daily Fama-French size and book-to-market factors⁸.

Using this idiosyncratic return volatility (VOLFF3), we use regression 2 to examine its effect on R&D investment. The results of this regression are reported in Table 9.

Consistent with the results in Tables 2 and 9 shows that VOLFF3 significantly affects future R&D investment. The coefficient for VOLFF3 in model 1 is 0.163, which is slightly higher than the coefficient reported in Table 2. The effect of VOLFF3 on R&D investment is still significant after we control for a firm's main characteristics (columns 4 and 5)⁹.

7. Conclusion

We examine the effects of uncertainty on R&D investment. Using the idiosyncratic volatility of stock returns to represent uncertainty, we find that firms invest more in R&D when they face higher uncertainty. This result holds when we control for several main characteristics of the firm and under different regression specifications.

⁸ We thank Ken French for making the data on SMB and HML available.

⁹ The results are also almost the same when we estimate idiosyncratic return volatility using the four-factor model. This test's results are available on request.

Table 9
Alternative idiosyncratic volatility and R&D investment.

	(1) $\text{RDAT}_{i,t+1}$	(2) $\text{RDAT}_{i,t+1}$	(3) $\text{RDAT}_{i,t+1}$	(4) $\text{RDAT}_{i,t+1}$	(5) $\text{RDAT}_{i,t+1}$
$\text{VOLFF3}_{i,t}$	0.163*** (0.000)	0.115*** (0.000)	0.134*** (0.000)	0.043*** (0.005)	0.066*** (0.000)
$Q_{i,t}$	0.009*** (0.000)	0.005*** (0.000)	0.003*** (0.000)	0.005*** (0.000)	0.003*** (0.000)
$\text{RDAT}_{i,t}$		0.348*** (0.000)	0.730*** (0.000)	0.343*** (0.000)	0.715*** (0.000)
$\text{CASH}_{i,t}$				−0.013*** (0.000)	0.021*** (0.000)
$\text{ROE}_{i,t}$				0.000 (0.598)	0.000 (0.592)
$\text{DIV}_{i,t}$				−0.013*** (0.000)	−0.012*** (0.000)
$\text{LAGE}_{i,t}$				0.005*** (0.000)	0.002*** (0.000)
$\text{LSALE}_{i,t}$				−0.008*** (0.000)	−0.002*** (0.000)
$\text{HHI}_{j,t}$				0.000 (0.923)	−0.002 (0.175)
Intercept	0.030*** (0.000)	0.018*** (0.000)	−0.000 (0.552)	0.048*** (0.000)	0.001 (0.347)
Fixed effects	F, T	F, T	Ind, T	F, T	Ind, T
N	90,650	90,650	90,650	90,650	90,650
Adj-R ²	0.1774	0.7571	0.7823	0.6856	0.7849

This table reports the results from the regression model of R&D investment on idiosyncratic volatility and other control variables: $\text{RDAT}_{i,t+1} = \beta_0 + \beta_1 Q_{i,t} + \beta_2 \text{VOLFF3}_{i,t} + \beta_3 \text{RDAT}_{i,t} + \beta_4 \text{FIRM_CHAR}_{i,t} + f_i + y_t + \varepsilon_{i,t}$ (2) where RDAT is the ratio of R&D expenditures to total assets, VOLFF3 is the standard deviation of residual stock returns using Fama and French's (1993) three-factor regression model, Q is the Tobin's Q that is the ratio of a firm's market value to its book value, VOL is the firm's idiosyncratic volatility, and FIRM_CHAR is a set of a firm's characteristics that potentially affect the firm's capital investment policies. f_i is a firm/industry dummy, and y_t is year dummy. All variables are defined in Appendix A. The sample period is from 1985 to 2013. Standard errors are clustered by firm, and p-values are reported in parentheses. The *, **, and *** denote statistical significance at 10%, 5%, and 1%, respectively. F denotes firm fixed effect, T denotes year fixed effect, and Ind represents industry fixed effect.

We further show that the effect of idiosyncratic return volatility on R&D investment is more pronounced for firms in more competitive industries. We also find that firms with product that have less market power such as small firms, as captured by sale volume, or young firms invest more in R&D when they face high uncertainty since these firms have greater pressure to survive competition.

To the best of our knowledge, this is the first empirical paper that uses the theory of the strategic growth option to investigate the relation between uncertainty and R&D investment. Consistent with the theory, we show that the preemptive strategy under competition is the main force driving the positive relation between uncertainty and R&D investment.

Appendix A. Variable definition and calculation

$\text{VOL}_{i,t}$	The idiosyncratic return volatility of firm i in year t , which is the standard deviation of residuals from the regression model of stock return on market returns over a year: $\hat{\sigma}_{i,t} = \sqrt{\frac{1}{t} \hat{\varepsilon}_{i,t}^2}$
$\text{VOLFF}_{i,t}$	The idiosyncratic return volatility of firm i in year t , which is the standard deviation of residuals from the standard deviation of the residuals from the Fama and French (1993) three-factor regression model.
$\text{RDAT}_{i,t+1}$	The ratio of R&D expenditures to total assets of firm i at the end of year $t + 1$ ($\text{XRD}_{t+1}/\text{AT}_t$)
$Q_{i,t}$	Tobin's Q , which is defined as the ratio of a firm's market value to its book value in year t ($(\text{MEF} + \text{AT-CEQ-TXDB})/\text{AT}$).
$\text{LOGQ}_{i,t}$	The logarithm of alternative measure of Tobin's Q , which is defined as the ratio of a firm's market value to its replacement cost of capital in

(continued on next page)

	year t $((MEF + AT-CEQ-TXDB)/K)$.
LOGK _{it}	The logarithm of a firm's replacement costs (K) of capital to the average replacement costs of capital for all firms in the same industries in year t . We follow the method of Salinger and Summers (1983) and use the perpetual inventory method to compute the replacement cost of the capital stock. We initialize the first value of capital stock (K_0) as gross PPE (PPEGT). We then construct the capital stock iteratively as $K_t = ((P_t/P_{t-1}) * K_{t-1} + CAPX_t)(1-\delta_j)$, where P is the price deflator for fixed nonresidential investment from national income and product accounts, CAPX is capital expenditure, and δ_j is book depreciation rate at the three-digit SIC level. We calculate $\delta_j = 2/L_j$, where L_j is the useful life of a capital good that is computed as $L_j = \frac{1}{N_j} \sum_{i \in j} \frac{PPE_{t-1} + I_t}{D_t}$ where D is depreciation (DP).
CASH _{it}	The ratio of cash holding to total assets for firm i in year t (CHE_t/AT_t)
ROE _{it}	The ratio of net income of firm i at the end of year t to its lag book equity $(NI_t/SEQ + TXDITC - PSTKRV - PRCA)_{t-1}$
DIV _{it}	The ratio of total dividends to total assets in year t $((DVC_t + DVP_t)/AT_t)$
LAGE _{it}	The logarithm of the firm's age in year t where the age is calculated from the CRSP data
LSALE _{it}	The logarithm of sales volume of firm i in year t ($\log(SALE_t)$)
HHI _{j,t}	The Herfindahl-Hirschman index of industry j in year t that is calculated from the sales volumes of the firms in Compustat $HHI(Sale)_{jt} = \sum_{i=1}^J s_{ijt}^2$ where s_{ijt} is the proportion of sales of firm i in industry j in year t , and J is the number of firms in industry j . Industries are defined by three-digit SIC codes.
HCON _{it}	The Herfindahl-Hirschman concentration index of a firm's sales across customers computed using data from the Compustat segment files. $HCON_{it} = \sum_p \left(\frac{Sales_{i,p,t}}{Sales_{i,t}} \right)^2$ where $Sales_{i,p,t}$ is the reported sales of firm i to customer p in year t , and $Sales_{i,t}$ is the total sales volume of firm i in year t .

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